

Dr. Babasaheb Ambedkar Technological University, Lonere
Regular and Supplementary - Summer 2024

Course: B. Tech. Branch: Civil Engineering Semester: VI

Subject Code & Name: BTCVC602_Y22 Foundation Engineering

Max Marks: 60

Date: 15/06/2024

Time: 3Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve <u>any two</u> of the following.		12
A) What are the purposes of soil exploration?	Remember	06
B) Draw the neat sketch of soil sampler showing its components. Explain the below terms associated with it: 1) Area Ratio, 2) Inside Clearance & 3) Outside clearance	Understand	06
C) A sampling tube has inner diameter of 80 mm and cutting edge of inner diameter 70 mm. it has outside diameter of sampling tube and cutting edge are 83 mm and 86 mm respectively. Determine the 1) Area Ratio, 2) Inside Clearance & Outside clearance of the sampler. The tube is pushed at the bottom of the bore hole to a distance of 600 mm with the length of sample collected as 550 mm. Find the recovery ratio and comment on the results.	Apply & analyze	06
Q.2 Solve <u>any two</u> following questions.		12
A) What are the general modes of bearing capacity failure of a footing. Explain in detail with sketches?	Understand	06
B) A strip footing of width 3 m is located at a depth of 1.40 m below the moist soil having cohesion $c = 20 \text{ kN/m}^2$ and angle of internal friction $\phi = 30^\circ$. Take FOS = 2.5, $\gamma = 18 \text{ kN/m}^3$, $\phi = 30^\circ$, $N_c = 37.2$, $N_q = 22.5$ and $N_\gamma = 19.7$. Determine (a) Ultimate bearing capacity, (b) Net safe bearing capacity, and (c) Gross safe bearing capacity Use the general shear failure theory of Terzaghi.	Apply	06
C) A strip footing of width 1.5 m resting on a saturated clay $\{C_u = 30 \text{ kN/m}^2, \phi_u = 0^\circ, \gamma_{sat} = 20 \text{ kN/m}^3\}$ is located at a depth of 2 m below ground level. The water table is also at a depth of 2 m from the ground level. Determine the ultimate bearing capacity of footing by using Terzaghi's Theory. If the water table rises by 1 m, calculate the percentage reduction in the ultimate bearing capacity.	Analyze	06

Q. 3	Solve <u>any two</u> of the following.		12
A)	What are expansive soils? Explain in detail the characteristics of expansive soils?	Understand	06
B)	Define and explain the following terms: A) Swelling Potential B) Swelling Pressure C) Free Swell	Remember	06
C)	Write a short note on settlement of shallow foundations?	Understand	06
Q.4	Solve <u>any two</u> of the following.		12
A)	State and explain in detail the Engineering News Record Formula (ENR) and Hiley's Formula?	Remember	06
B)	A precast concrete pile of size 30 cm x 30 cm is to be driven into clay strata whose unconfined compressive strength is 80 kN/m^2 . Compute the length of pile required to carry a safe working load of 400 kN with factor of safety of 2.5. Assume adhesion factor α as 0.6 and $N_c = 9.0$	Apply	06
C)	A 10 m long 300 mm diameter pile is driven in a uniform deposit of sand ($\phi = 38^\circ$). The water table is at great depth and is not likely to rise. The average dry unit weight of sand is 18 kN/m^3 . Calculate the safe load capacity of the pile with factor of safety of 2.5. Use $N_q = 137$	Analyze	06
Q. 5	Solve <u>any two</u> of the following.		12
A)	What are the different types of slopes. Explain in detail with neat sketch and example?	Understand	06
B)	Enlist and explain in detail the different types of slope failures?	Understand	06
C)	An infinitely long slope in dense sand having an inclination of 28° to the horizontal. Compute the factor of safety against shear failure if the angle of internal friction of the soil is 33°	Apply	06

*** End ***